

Data Management Plans (DMPs)

Introduction to Data Management Practices course

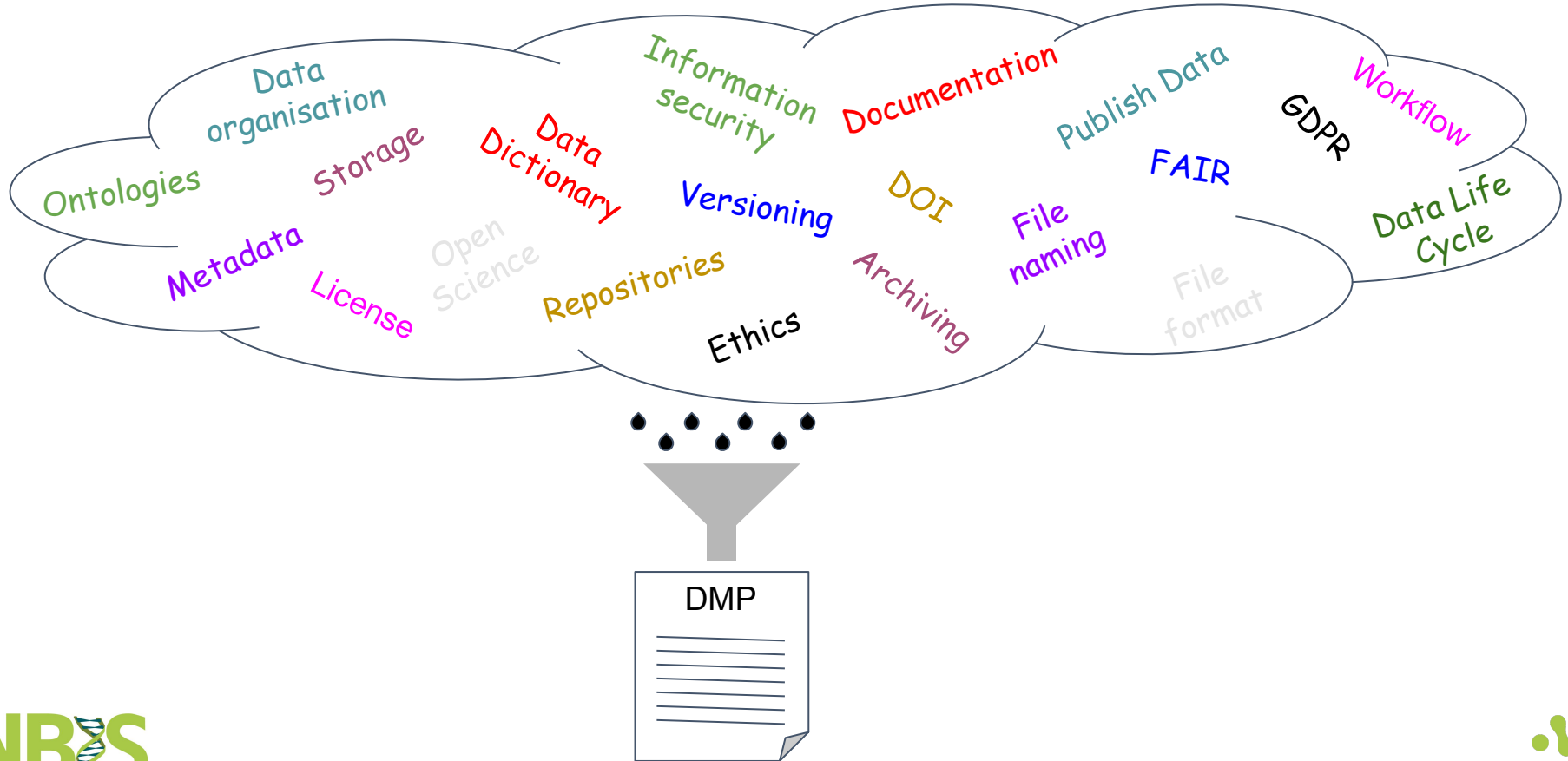
NBIS DM Team

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Learning objectives

1. Know the benefits of DMPs
2. Know the sections of a typical DMP
3. Know tools to assist in writing a DMP
4. Have experience in writing a DMP using an online tool called SciLifeLab Data Stewardship Wizard
5. Know where to get help in future DMP writing

Data Management - Now what?



What is a DMP?

A document **addressing requirements and practices** for managing the project's data, code and documentation **throughout the data life cycle**.

Outlines the data management **strategies**, and **how** the data is:

- collected
- documented
- organised
- preserved



Data Life Cycle by [RDMkit](#) used under [CC-BY](#)

Why should you write a DMP?

- **Identify gaps** in data management early
- **Define responsibilities**
- **Estimate costs**
- **Facilitate collaboration** by setting project wide standards (e.g. data organisation, naming conventions, metadata standards)
- Obtain **well-managed** research data (which enables reproducibility and reusability)
- First step towards **FAIRness**
- **Requirement** from funders and other stakeholders for reasons of **transparency, openness and return on investment**



Data Life Cycle by [RDMkit](#) used under [CC-BY](#)

When do you write a DMP?

A DMP is a **living document** that will develop throughout the project

- **Project planning:** Outline the strategies to be able to estimate the resources needed, so this can be included in the proposal for funding.
- **Project start:** Complete with details e.g. about documentation, data quality measures, file and folder strategies, etc.
- **Project end:** Update with e.g. links to published data and details about archiving (what data and where).

The main parts of a DMP

1. Description of data

- What types of data will be created and/or collected?

Formats

Amount/volume of data

Instrument

Equipment

2. Documentation

- How will the material be documented and described?

ELN & LIMS

Collection method

Metadata standards

Versioning

The main parts of a DMP

3. Storage and backup

- How is data security, storage and backup handled?

Organization

Naming convention

Backup strategy

Security

4. Legal and ethical aspects

- How is data handled? Any legal requirements?

Sensitive data

GDPR

Intellectual property rights

Nagoya

The main parts of a DMP

5. Accessibility and long-term storage

- How, when, and where will research data and metadata shared?

Repositories

Raw data

Code & Software

Type of storage

6. Responsibility and resources

- Who are the responsible persons for data management?

Resources

Labour estimate

How write a DMP?

DMP templates:

- Provided by funding agencies, e.g. [Swedish Research Council](#) and [Science Europe](#)
- High-level questions, with no guidance on how to answer
- Use MS Word?

DMP tools:

- [DMPOnline](#) - Common for universities; Good guidance but not Life Science specific; Free text answer type;
- [Data Stewardship Wizard](#) - Provided by [SciLifeLab](#); Life Science specific guidance; Many questions with answer options



Key Points

- A data management plan (DMP) is a document that describes the data produced in the course of a research project
- A DMP allows for well-managed data, and funding agencies often require a DMP for transparency and return on investment
- A DMP is a living document, the first version is written during project planning, and is then updated as the project proceeds
- There are standard templates available e.g. at funder agencies, and tools to assist when writing

Exercise: Write a DMP

- Using the scenario listed in Canvas, create a DMP using the DSW
- Follow the instructions in Canvas to login to DSW and to create a new project
- Fill in the information section by section
- Finish by creating a PDF of your DMP

Discuss your experience

1. Have you learned something today that you wish you knew earlier?
2. What is the biggest challenge you foresee in implementing a data management plan?
3. What is one question you still have about data management plans?